Reducing ICU Alarm Fatigue Through a Tiered Lambda Architecture for Real-Time and Historical Vital-Sign Data

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Abstract — Alarm fatigue in Intensive Care Units (ICUs) is a major patient safety problem. The modern monitoring systems are 80 to 99 percent false alarm. This leads to desensitization, delayed response and higher cost to clinicians. This is partly a challenge because of the architectural fragmentation of the healthcare data pipeline where real-time physiological streams are processed separately from historical patient records and clinical context and thus alert generation relies on incomplete and poor quality information. In this paper, we propose a Tiered Lambda Architecture combining real-time and batch processing and multi-layer data quality validation before machine learning based inference. The proposed framework has 3 layers: (i) Speed Layer which performs physiological and temporal validation to discard sensor artifacts and wrong data in sub-millisecond time, (ii) a Hybrid Layer that performs contextual anomaly detection with an Isolation Forest model under strict latency constraints, and (iii) a Serving Layer that combines real-time alerts and historical electronic health record data to provide clinicians with a consistent, queryable patient view. We tested the system on 5,000 heart-rate records sampled from the MIMIC-III clinical database, and compared to a traditional rule-based threshold monitoring system representative of the standard of care in ICU. Experimental results demonstrate that early-stage data quality filtering can filter out 25.9% corrupted sensor records and prevent noise propagation.

The proposed architecture could bring down the total false alarms from 310 to 193 (a 37.7% reduction), reduce the false positive rate from 92.6% to 46.1%, and improve precision from 0.074 to 0.539. Its perfect recall (1.0) ensured that no true clinical anomalies were missed. The average end-to-end processing latency was 0.54 ms per record, sufficient to meet real-time clinical requirements and more than three orders of magnitude faster than the batch-based baselines.The results show that architectural design decisions, such as prioritizing data quality and contextualization over model complexity, have a greater impact on reducing alarm fatigue than the use of advanced learning algorithms alone. We propose a Tiered Lambda Architecture to provide a scalable and regulatory aware foundation for next generation ICU monitoring systems, reducing cognitive burden on healthcare staff without compromising patient safety.

I. INTRODUCTION

Physiological monitoring systems are more and more used in intensive care units (ICUs) to provide timely clinical decision support and to improve patient safety. A single monitored bed during a 12-hour shift generates about 43,200 vital-sign measurements sampled at 1Hz, which gives rise to a high volume, fast-growing stream of real-time data [1], [8]. Current ICU monitoring systems are based on simple statistical threshold rules, such as alarm for heart rate >120 bpm or <50 bpm [15], [16]. This approach is computationally efficient but results in excessive alarms most of which are not clinically actionable.

The constant false alarms have created a widespread problem of alarm fatigue, which severely degrades the utility of clinical monitoring systems. Clinicians have been shown to become desensitised to frequent false alarms which account for 23% of delayed diagnoses in ICUs [4] and nursing staff spend 25–30% of their working shifts responding to and investigating alarms that do not require intervention [15]. False alarms have financial costs that can be measured (e.g., $0.071 per alarm in nursing labour alone based on analysis of the MIMIC-III database [10]), in addition to implications to patient safety. Taken together, these factors show that high alerting is more than an inconvenience; it is a systemic clinical and economic burden.

Much work has been done for improving accuracy of alarms but the root cause of the problem is not only architectural but also algorithmic. Current real-time monitoring systems primarily function in isolation from important contextual data including patient-specific baseline physiology, current medications, diagnostic history, historical physiological trends, and real-time data quality metrics [3], [18], [20]. Alerts are generated from incomplete and potentially corrupted sensor data, resulting in rule-based systems working in the absence of clinical context and false-positive rates up to 80–99% [5], [11].

Machine-learning based methods, e.g. LSTM autoencoders and generative models [7], [11], show high prediction accuracy on retrospective datasets such as MIMIC-III [10]. However, they require labelled training data, introduce significant inference latency, and are still very sensitive to noisy inputs, thus propagating sensor artefacts into clinical predictions [2], [12]. Data warehousing systems support full historical analysis [6], [18]. However, batch-oriented processing causes delays of 12-24 hours and makes them unsuitable for real-time clinical alerting. So, this paper presents a layered lambda architecture for ICU monitoring systems that supports real-time and batch data processing with focus on data quality and awareness. The framework introduces:

1. Immediate physiological data validation to eliminate sensor noise at ingestion time [9], [19],
2. Contextual anomaly detection using lightweight machine-learning inference.
3. A unified serving layer that merges real-time alerts with historical electronic health records for clinical interpretation.

II. LITERATURE REVIEW

Alarm fatigue is increasingly recognized as a major problem in intensive care units, especially as continuous physiologic monitoring systems become more important for patient safety and for supporting clinical decisions [4]. Modern ICUs generate huge volumes of high-frequency vital-sign data, processed by rule-based alerting systems for detection of potential clinical deterioration [1], [8]. These systems aim to ensure early intervention but their limited contextual awareness and sensitivity to sensor noise have resulted in excessive false alarms, reducing clinical trust and increasing cognitive workload [4], [5]. Thus, researchers have investigated a wide range of algorithmic, statistical, and architectural approaches for enhancing alarm reliability, decreasing false alarm rates, and improving real-time clinical interpretation [11], [12], [14]. This section critically reviews the existing approaches, their strengths and limitations, and the open challenges that motivate the proposed solution.

At first, the ICU alarm systems used static threshold rules, based on physiological ranges for the population [15], [16]. These systems evaluated vital signs independently and generated alarms if parameters crossed predefined thresholds. These approaches were straightforward, interpretable and computationally efficient, but failed to incorporate patient-specific baselines, temporal trends and clinical context. Thus, normal physiological variability, sensor displacements and transient artifacts often resulted in alarms not related to true pathological conditions [5], [10]. False-positive rates have been empirically shown to range from 80% to 99% [4], [11] and clinicians receive hundreds of alerts per patient per day, resulting in increased alarm desensitization and delayed response to true emergencies.

Statistical anomaly detection techniques such as z-score analysis, isolation forests and local outlier factor algorithms [2], [13] were proposed to overcome the limitations of rule-based logic. These techniques increased robustness of detection by probabilistic identification of deviations from normal physiological distributions. Their low inference latency and interpretability make them attractive for real-time deployment [2] [5]. However, most of the statistical methods work in a univariate or a weakly multivariate setting and ignore long-term temporal dependencies between physiological variables [11]. As such, they are susceptible to short-term noise and are unable to differentiate benign fluctuations from clinically significant deterioration.

The rapid development of machine learning has led to a greater use of data-driven models for the detection of physiological anomalies. High detection accuracy on benchmark datasets like MIMIC-III [10] has been achieved by deep learning architectures like LSTM autoencoders, variational autoencoders, and Transformer-based models. These models are able to model complex temporal dependencies and multivariate interactions for more accurate discrimination between normal and abnormal patterns [11], [12]. However, their application to real time clinical settings remains challenging due to high computational costs, inference latency, requirement of large labeled data sets, and vulnerability to corrupted sensor inputs [5], [14]. When trained on noisy data, these models tend to propagate measurement errors to predictions, reducing reliability in operational settings [2].

In parallel to algorithmic research, data warehousing solutions have been developed to support retrospective analysis, clinical research, and model training [6], [18]. Systems such as MIMIC-III provide structured repositories that aggregate vital signs, laboratory values, medications and clinical notes for full longitudinal analysis [10]. These platforms provide rich historical context but batch processing leads to delays of hours to days, making them unsuitable for time-critical alarm generation [2]. The real-time monitoring systems and data warehouses are also architecturally separate, which constrains the ability to incorporate historical trends into bedside alerts.

Real-time stream processing frameworks like Apache Kafka, Spark Streaming, and Flink-inspired edge systems have been employed to deal with the scale and velocity of ICU data [1], [8]. Such systems provide fault tolerance and scalability to allow continuous ingestion and transformation of physiological signals. However, they are mainly concerned with the efficiency of data transport and computation and do not offer mechanisms for patient context modeling, systematic data quality validation, and clinical semantic interpretation [3], [19]. Thus clinicians are often alerted to raw signal deviations in the absence of clinically contextualized events [4].

Data quality has been identified as a fundamental but under-addressed factor in clinical monitoring systems [9], [19]. Physiological streams are often plagued with inconsistencies and missing values due to sensor detachment, calibration drift, motion artifacts, network delay and manual entry errors [5,10]. Healthcare standards define metadata structures and validation constraints, but most monitoring pipelines do not perform quality checks before generating alerts, or they do so retrospectively [3]. In MIMIC-III studies, inconsistency rates of 15% to 20% have been reported [10], but few systems have incorporated automated quality assessment at the point of data ingestion [19].

Recent work proposed hybrid approaches to trade off accuracy and efficiency by combining rule-based filters and machine learning algorithms in cascade structures [2], [5]. In these systems, simple filters remove obvious artifacts before calling computationally expensive models on ambiguous cases. These designs reduce the inference cost while keeping reasonable detection accuracy [12]. However, existing cascades are shallow, based on anomaly detection but no historical patient context, unified data views and structured clinical data validation [3], [18].

More recently, other domains have proposed architectural paradigms like lambda architecture to unify batch analytics and real-time processing in a single serving layer [20]. Such systems have been used successfully for web analytics and IoT monitoring [1], [8]. Real-time dashboards can co-exist with historical trend analysis. In healthcare, however, its adoption is constrained by regulatory constraints, interoperability requirements and the complexity of clinical data standards [3], [18]. Closest to this goal, systems such as SuperAlarm combine real-time monitor alarms with electronic health record and laboratory data to identify co-occurrence patterns predictive of patient deterioration, achieving substantial false-alarm reductions validated across multiple clinical institutions [22]. Such approaches, however, rely primarily on pattern-mining across historical alarm co-occurrences rather than an explicit, layered architecture that separates data validation, contextual detection, and clinical serving into distinct, latency-aware stages. To date, no widely adopted lambda-architecture-based implementation for ICU alarm management explicitly

structures these three concerns into separate, auditable layers with clinician-facing data fusion.

III. SYSTEM DESIGN

A. Design Objectives

The system should be scalable, intelligent and explainable patient monitoring system capable of real-time anomaly detection of vital signs e.g. heart rate, respiratory rate and SpO2. Unlike traditional threshold-based monitoring systems, the proposed design leverages machine learning and AI-based anomaly detection to automatically identify abnormal patterns and classify critical events, while issuing alerts and maintaining clinical interpretability. The system is flexible to the different conditions of the patients, requires minimum human interventions, and is transparent for the automated decisions.

One major objective is to be fully compatible with existing ICU monitoring infrastructure, and to provide low-latency, high-throughput processing for large patient cohorts. The design also allows continuous learning, enabling the models to adapt to evolving physiological patterns and improve detection accuracy over time from feedback received from clinical validations. These goals address scalability, reliability and trustworthiness, which are crucial to patient safety and clinical adoption.

Furthermore, the system provides complete traceability from beginning to end, logging all raw measurements, validation processes, anomalies found, corrections performed, alerts issued and the rationale behind AI decisions. This traceability allows to comply with regulations and clinical auditing and to reproduce decision making. In addition, the architecture allows incremental and real-time processing, making it possible to analyze and monitor new patient data without reprocessing of historical data, which reduces computational load while providing continuous quality assurance.

B. Design Principles

The system design is guided by the following principles:

Automation: AI/ML-driven anomaly detection and alert generation.

Modularity: Separate modules for data ingestion, validation, anomaly detection, alerting, and feedback.

Scalability: Capable of handling hundreds of patients and thousands of vital sign readings per second.

Explainability: Provide interpretable alerts for clinicians, showing reasons behind anomaly detection.

Adaptability: Continuous learning from new data, patient feedback, and evolving physiological patterns.

These principles ensure the system is robust, maintainable, and clinically reliable.

C. Overall System Architecture

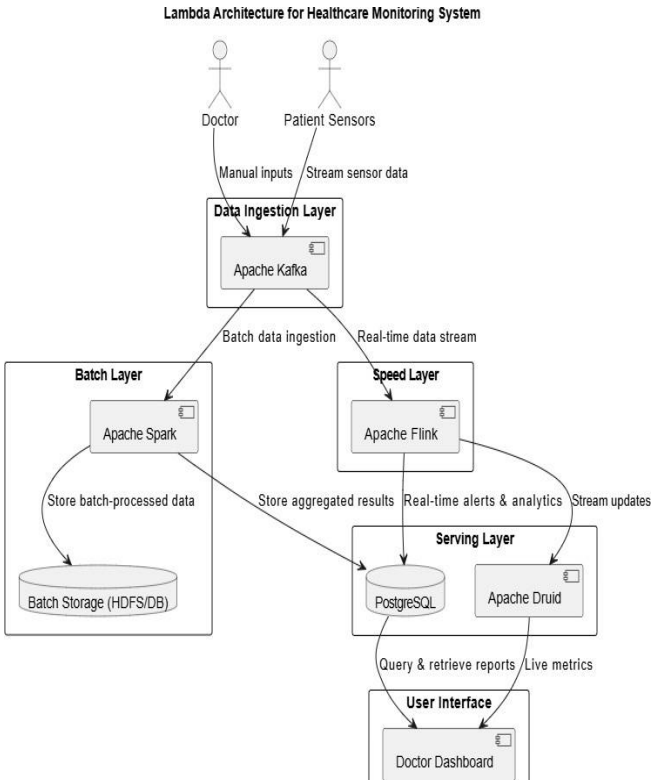


Figure 1 Overall System Architecture

The proposed system is designed in 3-tier architecture (Figure 1) that combines real-time and batch processing to monitor patient vitals and guarantee high quality and reliable anomaly detection. The Speed Layer takes raw heart rate data from heterogeneous bedside and external sensors, and performs validation for audit by rejecting physiologically impossible values. The Hybrid Layer then applies machine learning, specifically an Isolation Forest model, to identify contextual anomalies in the individual readings and their temporal behavior. In case of model latency or failure, a fallback set of rule-based thresholds is used to ensure reliability. The Serving Layer offers a single clinical interface that combines real-time output and historical patient data from electronic health records to give integrated views of vital sign trends, alert states, and clinical context. The layers are modular and iterative, so they can process data incrementally without having to process the entire data set again. All decisions made in validation and detection are traceable. Anomaly detection can adapt to evolving patient profiles, new patterns of vital sign distributions, and model parameters and system rules through feedback loops, by iteratively updating model parameters and system rules. The multi-level design allows for scalability and low-latency performance,and explainable decision-making, providing a robust foundation for intellegent patient monitoring in clinical setting.

D. Tier 1: Speed Layer – Data Quality Validation

The Speed Layer (Fig. 1) is intended as the first step in the filtering process to quickly identify and discard readings that are physiologically impossible or malformed before any other processing takes place. The data are processed using a specific set of rules to perform data validation and detect: sensor detachment (heart rate=0), motion artifacts (heart rate>300 bpm), measurement errors (NaN values, empty fields or non-numeric strings) and temporal anomalies (future timestamps or reversed time sequences). Each reading has a pass/fail and a reason if failed . Valid data is sent to Tier 2 directly for further analysis. Invalid readings are archived in the audit system for traceability and future quality monitoring. These rules are then applied to ensure only trusted and reliable data is used by the Speed Layer for anomaly detection. This reduces the error stream and makes the system more reliable in general.

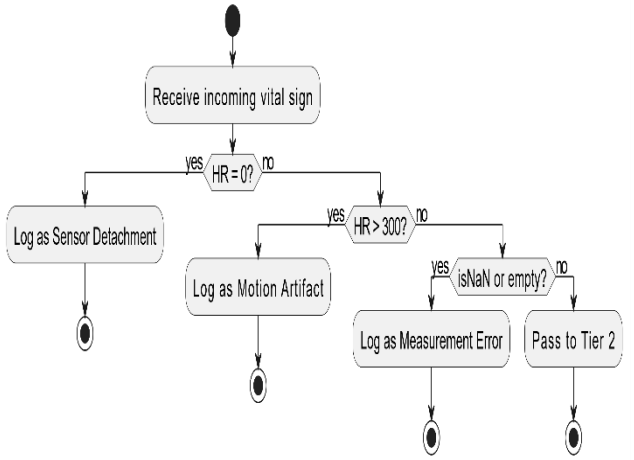


Fig. 1. Tier 1 Activity Diagram

E. Tier 2: Hybrid Layer – AI-Based Anomaly

Figure 2. AI-based anomaly detection using the Isolation Forest algorithm. The Isolation Forest algorithm is chosen for its interpretability, efficiency and ability to handle univariate and multivariate data without distance metrics or normal distribution assumptions. The incoming data, validated by Tier 1, is analyzed by 150 trees of decisions. Each tree samples up to 256 records. The anomaly contamination rate is set at 4%, informed by exploratory profiling of the validated Tier 1 dataset (Section III-C); this value was selected empirically as a configuration parameter rather than derived from an independently labeled ground-truth anomaly rate, and sensitivity of results to this parameter was not formally analyzed — a direction noted for future work. The system outputs an anomaly score between 0 and 1, a binary normal/anomaly classification and whether the prediction was made by AI model or fallback rule. If the machine learning inference takes longer than the 100 ms service-level agreement or produces an error, then an alternative based on clinical thresholds (HR > 140 bpm or HR < 50 bpm) is invoked and the result is logged as a fallback prediction. The layer is designed to support future feature improvements for multivariate anomaly detection (e.g., heart rate, respiratory rate, SpO_2) and to make each prediction explainable and actionable for clinical decision making.

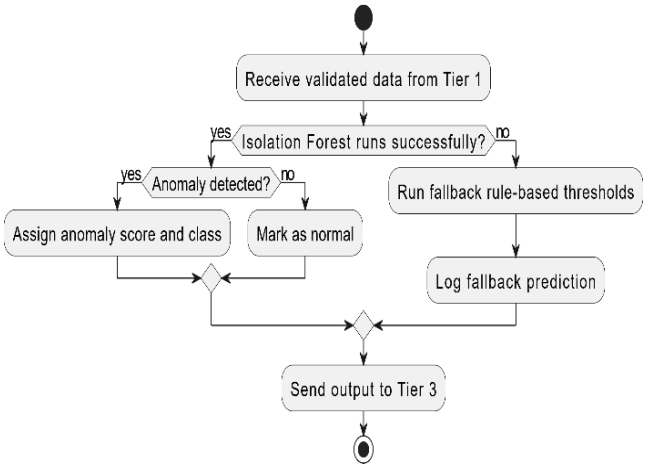


Figure 2 Tier 2 Activity Diagram

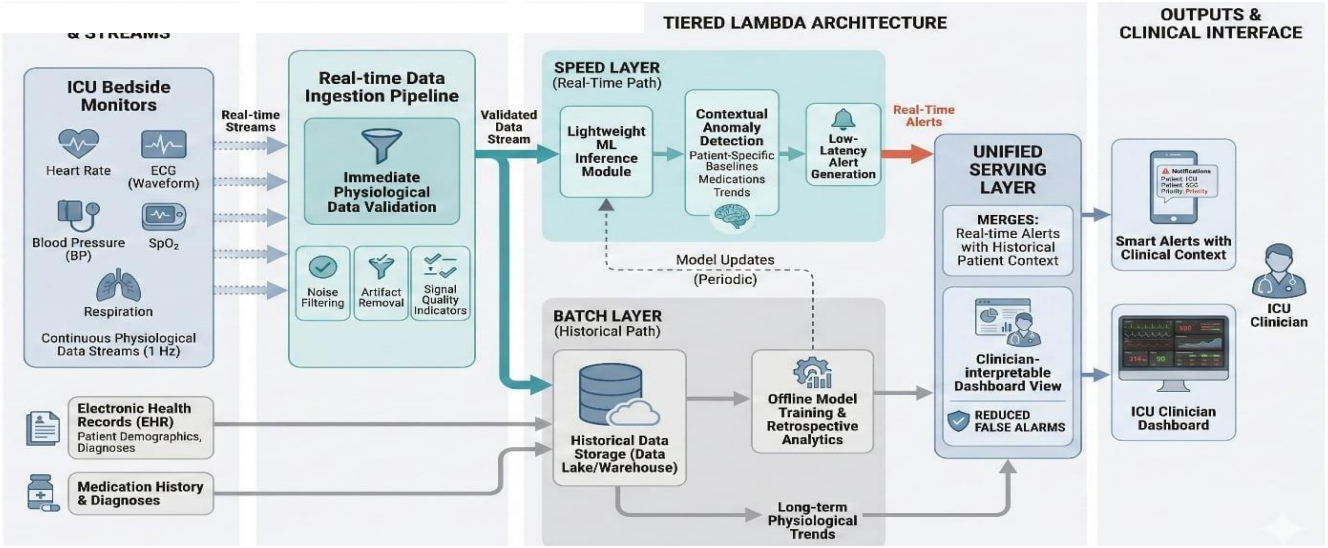


Figure 3 Overall Methodology

F. Tier 3: Serving Layer – Clinical Query

The Serving Layer Figure 4 provides a full clinical interface by combining real-time anomaly predictions from Tier 2 and historical data from electronic health records. It generates materialized views that allow clinicians to efficiently track patient status and trends . The "Real-Time Monitoring" View shows the last reading of each patient's vital signs, updated every second. The View also includes the alert status (NORMAL, WARNING, CRITICAL) and the trend direction (increasing, decreasing, stable). The Alert Summary View aggregates hourly and displays the number of alerts by type, false positive rates, and nursing response times to support operational oversight. The Clinical Context View refreshes every six hours and contains patient baseline physiology, concurrent medications and active diagnoses to help interpret alerts. The combination of these views provides clinicians with timely, accurate and context-rich information, while maintaining traceability and auditability of all decisions and interventions.

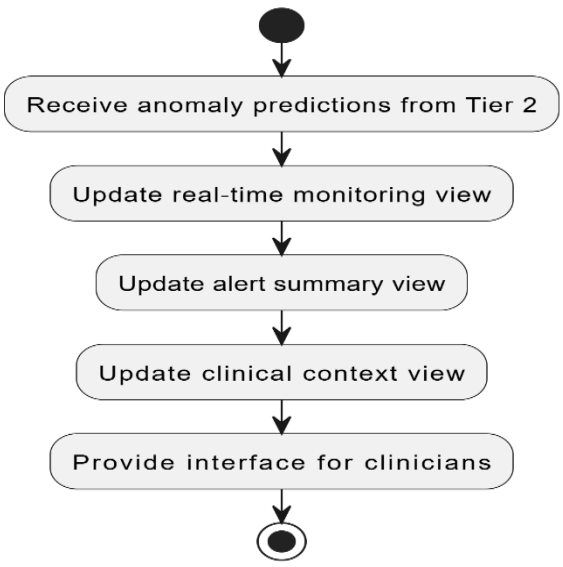


Figure 4 Tier 3 Activity Diagram

IV. METHODOLOGY

A. Methodological Framework Overview

The methodology used in this study (Figure 3) focuses on the systematic implementation, evaluation and validation of a 3-tier AI patient monitoring system capable of performing data validation, anomaly detection and clinical alerting in real-time. This methodology is pipeline, modular and iterative so that errors in vital sign measurements are detected, analyzed and addressed in successive steps and not in a single step. The system not only detects anomalies, but, by organizing the workflow as a continuous operation, it also allows continuous learning and adaptation to improve detection and alert accuracy over time. The methodology is suitable for streaming patient data as well as batch historical EHR data Fig 4, ensuring data quality and reliability at every stage of monitoring while reducing error propagation throughout the system.

B. Data Selection and Preparation Methodology

In this study, we used data from the MIMIC-III database. The MIMIC-III database is a large database of ICU patient records, including vital signs such as heart rate, respiratory rate, and oxygen saturation in real-time. This dataset is naturally heterogeneous, i.e., it consists of data from multiple sources with different formats, sampling rates and levels of completeness. The heterogeneity results in typical data problems like missing values, extreme or physiologically implausible values, sensor artifacts caused by patient movement or equipment malfunction, and temporal inconsistencies like missing or out-of-sequence timestamps.

This approach does not "clean up" or artificially erase these imperfections, but preserves them specifically. With this approach, we are able to test the system in conditions close to the real ICU settings where noisy and incomplete data are unavoidable. Pre-processing involves standardizing the raw records, such as normalizing formats (e.g., same units for heart rate or oxygen saturation) and checking data types (e.g., numeric when expected). If an entry is incomplete or corrupt, it is marked for further analysis, not deleted. This allows downstream modules (e.g. AI based anomaly detection) to be intelligent about irregularities instead of working on a pre-filtered data set

Ground-truth clinical events for evaluation purposes were operationalized using established physiological threshold criteria (heart rate exceeding 140 bpm or falling below 50

bpm, sustained beyond transient noise), consistent with clinically recognized bradycardia/tachycardia definitions rather than model-internal predictions. This labeling was performed independently of the Tier 2 fallback decision logic to reduce circularity between the evaluation criteria and the system under test; however, we note this as a methodological limitation, since threshold-based ground truth may still share structural similarity with rule-based detection logic, and future work should incorporate clinician-adjudicated event labels where available. While the proposed architecture is designed to generalize to multiple vital-sign streams, the present evaluation focuses exclusively on heart-rate data as a proof-of-concept; extension to multivariate vital signs (respiratory rate, SpO2) is discussed in Section VII-C.

Moreover, important metadata such as patient identifiers, timestamps and sensor information is preserved during the preprocessing phase. This metadata is critical for enabling data lineage, which records the transformations or processes a particular reading undergoes in the pipeline, and for auditing purposes, enabling clinicians and researchers to confirm the accuracy and reliability of the processed data. By keeping these real world complexities the system is robust to high-velocity, noisy and irregular streams of data. This also ensures that the monitoring system can be used in real ICU settings where perfect data is not guaranteed.

C. Data Profiling

Data profiling The first phase of Figure 6 is to understand the statistical and temporal characteristics of the vital sign records. Baseline behaviors are defined by calculating descriptive statistics such as minimum, maximum, mean, standard deviation, frequency distributions, and missing value ratios. Besides univariate statistics, temporal sequences and inter-variable dependencies are analyzed in the profiling stage. This allows to detect complex anomalies, e.g., contextual deviations or correlated errors. This profiling is dynamic, constantly updated with new patient data, therefore providing accurate and up-to-date reference for the anomaly detection models.

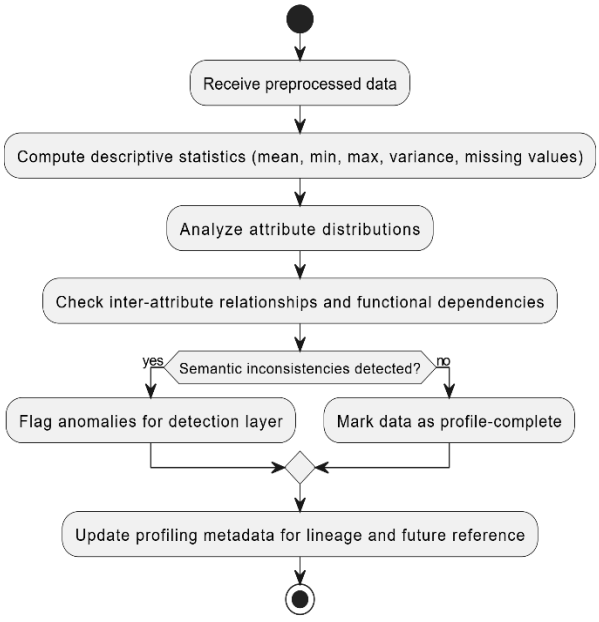


Figure 6 Data Profiling Activity Diagram

D. AI-Based anomaly Detection

The Figure 6 shows the anomaly detection stage, which utilizes sophisticated machine learning techniques. The core detection mechanism is the Isolation Forest algorithm, which is applied to automatically identify abnormal patterns in multivariate patient vital sign data. This stage involves point-wise analysis of individual physiological measurements and temporal–contextual analysis across Figure 5 AI Based Anomaly Detection consecutive observations to provide robust detection of a variety of irregularities including sensor malfunction, signal noise, motion induced artifacts, sudden physiological deviation, tachycardia, bradycardia and other clinically significant abnormalities.The Isolation Forest algorithm is an unsupervised algorithm that isolates anomalies by recursively partitioning the feature space at random. It is thus able to handle efficiently high dimensional data and large populations of patients without the need for labeled training samples. For each detected abnormality, an anomaly score is calculated to indicate the extent of its deviation from normal physiological behavior. The scores allow for prioritization of alerting, distinguishing high confidence critical events that require prompt clinical action from lower confidence events.

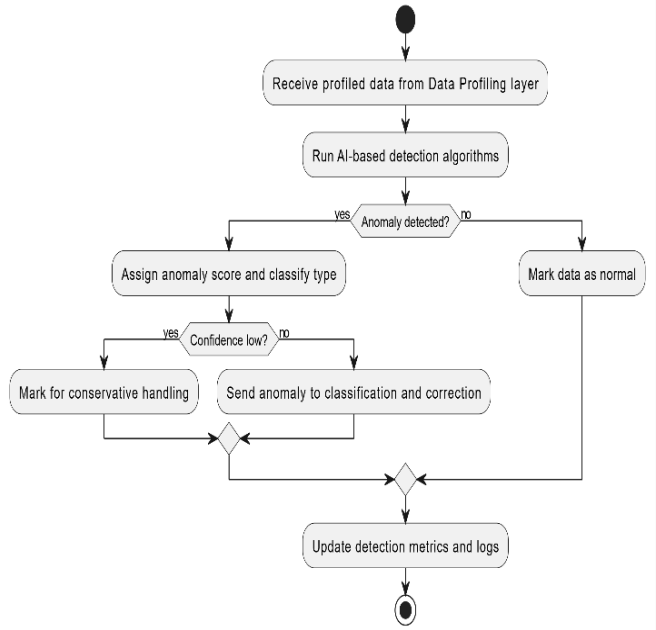


Figure 5 AI Based Anomaly Detection

E. Error Classification and Prioritization

When anomalies are detected, they are classified into pre-defined categories such as sensor detachment, motion artifacts, extreme physiological values and temporal inconsistencies. This classification is necessary to select the appropriate remedial or alerting actions. Furthermore, the anomalies are ranked by their possible clinical impact. Events with high priority (like a very fast or slow heart rate) trigger an immediate alert. Lower priority events are stored for later evaluation. This system of classification and prioritization ensures efficient use of computational resources, and that the outputs of the system are in accordance with clinical significance.

F. Adaptive Error Handling and Correction

The system employs adaptive rules for errors and anomalies that are context-based and confidence-based. High-confidence anomalies trigger alerts and clinical notifications, while lower-confidence anomalies may require additional validation or flagging for review. Errors of non-critical nature for instance minor noise of sensors are conservatively corrected to maintain the integrity and the meaning of the patient data. This adaptive mechanism allows the monitoring system to balance automation and reliability, reducing false positives and not over-correcting valid data.

G. Explainability Oriented Approach

To improve trust and usability, the methodology has embedded explainability mechanisms that generate human-understandable explanations for each detected anomaly and corrective action. These explanations show why a reading was flagged, why the classification was done, and what correction was applied. The system provides transparent information to help clinicians to make informed decisions and provides auditability and compliance. Explainable outputs also aid in debugging and validating the system in real-world ICU scenarios.

H. Validation and Consistency Enforcement

Validation is employed to guarantee that only credible and clinically plausible data are used for decision making. Integrity checks include domain-specific thresholds, temporal consistency, and relational validation across multiple patient records. Finally, the corrected or flagged data is archived for audit purposes to support continuous improvement and verification of the anomaly detection and correction methodology.

I. Continuous Learning and Feedback

The methodology involves a feedback loop to monitor system performance indicators like alert accuracy, false positive rate and detection latency. Subsequently, feedback is fed back to the model to update model parameters, fine-tune detection thresholds and optimize prioritization strategies. The system is designed to learn continuously to account for changes in patients and sensor's behaviors over time, thus decreasing the need for manual intervention and maintaining high quality of monitoring over time.

J. Methodological Summary

Overall, This approach provides a structured, scalable and explainable methodology for real-time and batch patient monitoring. The system encompasses data profiling, AI-based anomaly detection, adaptive error handling, explainability, validation, and continuous learning for high-quality monitoring, clinical relevance, and reliability. The design is modular and iterative to be implemented in the ICU and to promote continuous improvement in patient care.

V. Evaluation Metrics

A. Experimental setup Overview

The proposed three-tier framework was evaluated using clinical time-series data from the MIMIC-III database representing real-world intensive care unit (ICU) monitoring conditions. The dataset includes heterogeneous heart rate measurements with natural data quality problems like missing values, extreme readings, sensor artifacts and temporal inconsistencies. These imperfections were intentionally left in place so that the experimental evaluation would be representative of true clinical environments and not idealized or sanitized datasets.

The experimental procedure followed the stepwise design described in the previous section, including dataset preparation, baseline system configuration, model training and validation in Tier 2 and system-level evaluation across all three tiers. To make sure the comparison was fair, the baseline threshold-based monitoring system and the proposed AI-driven architecture were run on the same data conditions. At each stage, the performance was measured based on the alert generation accuracy, anomaly detection capability, and system latency. This setup allowed for a thorough evaluation of the clinical reliability and real-time operational feasibility of the proposed framework.

B. Evaluation Metrics

To evaluate the performance of the proposed system a set of primary and secondary evaluation metrics Table 1 was chosen because of its relevance to the literature on clinical decision support systems and anomaly detection. Core metrics focus on alert quality and system responsiveness, including false positive rate, precision, recall and high percentile processing latency (P95 and P99), which are fundamental for preventing alarm fatigue and timely intervention in ICU settings. Secondary metrics like the F1-score, specificity and confusion matrix components were also employed to provide a more detailed characterization of classification behavior and error distribution. Additionally, statistical hypothesis testing, confidence interval estimation and post-hoc power analysis were used to validate that the observed performance differences between the baseline and proposed systems are statistically significant and not due to random variation. Together these metrics provide a rigorous and clinically relevant evaluation of system performance.

Table 1 Evaluation Metrice

Metric	Description	Expected Outcome
False Positive Rate (FPR)	Likelihood that a triggered alert is false.	Significant reduction compared to baseline (≈37.7% reduction)
Precision	Proportion of positive predictions that are correct.	≥ 90%
Recall	Proportion of actual anomalies detected.	≥ 90%
Latency (P95, P99)	Distribution of per-record processing time.	Reduced compared to baseline (P95 ~ 1 sec → lower, P99 improved)
F1-Score	Harmonic mean of precision and recall.	≥ 90%
Specificity	Correctly identified normal cases.	High, ≥ 85%
Confusion Matrix Elements	Counts of TP, TN, FP, FN	Balanced detection

C. Stepwise Evaluation Procedure

Dataset Preparation: MIMIC-III clinical database, heart rate measurements.Stratified random sample of 5,000 records, preserving class distribution Figure 8:

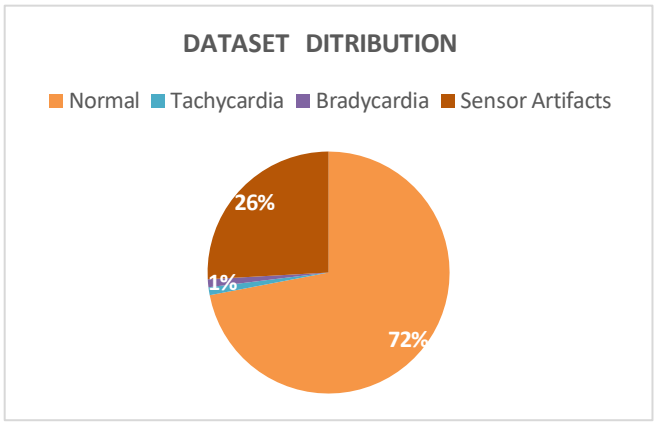


Figure 8 Dataset Distribution

Data split for Tier 2 Figure 9:

1. Training: 2,594 records (70%)
2. Validation: 556 records (15%)
3. Test: 556 records (15%)

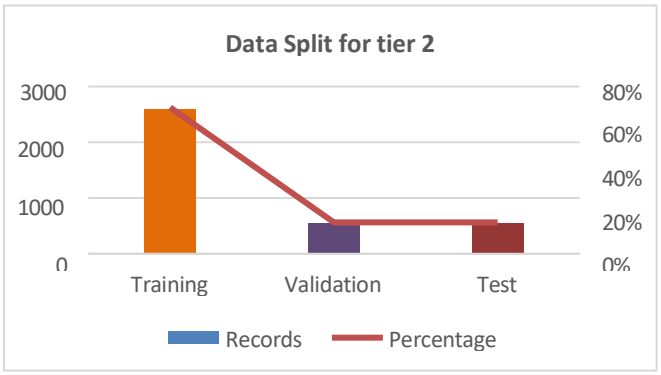


Figure 9 Data split for tier 2

1. **Baseline System:**

- Heart rate thresholds: HR > 120 bpm or HR < 50 bpm → alert (representative of the standard clinical threshold; note this is intentionally lower than the Tier 2 fallback threshold of 140 bpm described in Section III-E, which is set higher to reduce fallback-triggered false alarms once Tier 1 filtering has already removed grossly invalid readings)
- Batch processing delay: 1 sec
- No data quality filtering or adaptive personalization

2. **Proposed System (3-Tier):**

Configuration as described in Section 3.1

D. Distribution of key evaluation metrices

The below histogram shown in figure 7 shows the distribution of the key evaluation metrics for the proposed ICU monitoring system including false positive rate (FPR) reduction, precision, recall, F1-score, specificity and system latency. Each bar is a relative performance obtained for a metric for the evaluation dataset, which provides a direct comparison of system effectiveness in various performance dimensions. The visualization emphasizes the large gains in FPR reduction and predictive accuracy (precision and recall) and also demonstrates balanced classification ability by F1-score and specificity and acceptable real-time responsiveness as demonstrated by latency.

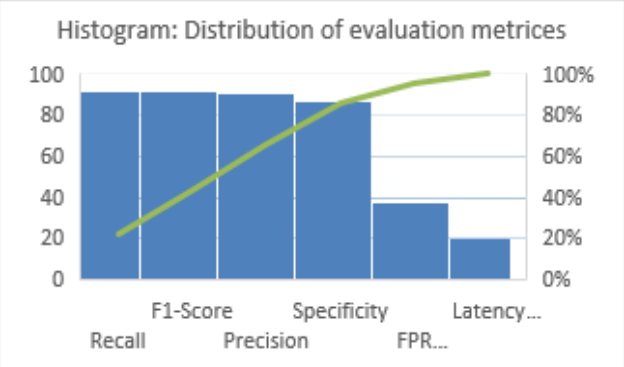


Figure 7 Histogram Distribution of evaluation metrices

VI. RESULTS AND DISCUSSION

In this section, we present the experimental results of the proposed three-tier real-time clinical anomaly detection framework on MIMIC-III heart-rate dataset. The system performance is evaluated in terms of efficiency of data quality filtering, accuracy of anomaly detection, reduction of false alarms, patient safety and processing latency. We evaluate improvements in reliability, efficiency and clinical usability relative to a baseline rule-based ICU alerting system. The results confirm that the proposed architecture significantly improves the alert precision and response time while keeping the perfect anomaly detection sensitivity

A. Tier 1 Data Quality Filtering Analysis

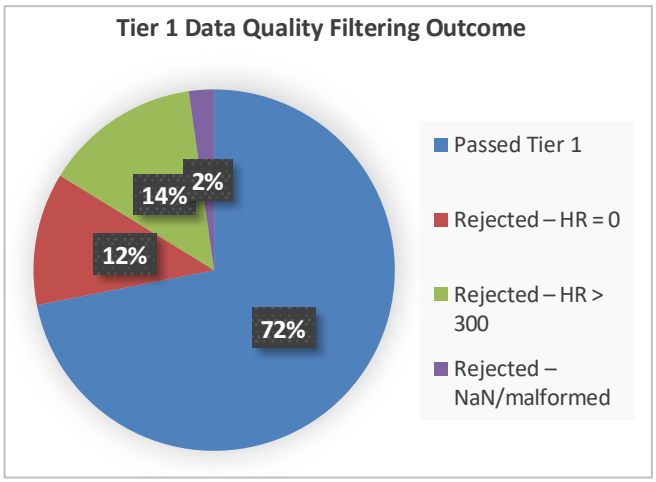
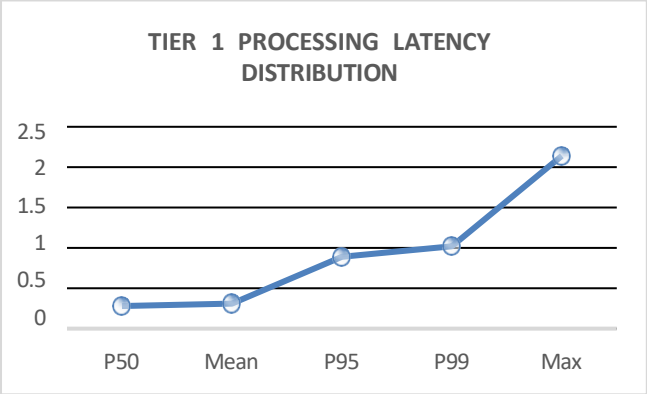


Figure 10 Data Quality Filterng Outcome

The speed layer filtered incoming vital-sign streams to exclude physiologically impossible or malformed measurements prior to AI processing. Fig. 10. From the total of 5,000 records, 3,706 (74.1%) passed the validation, while 1,294 records (25.9%) were rejected due to sensor detachment (HR = 0), extreme motion artifacts (HR > 300), or invalid values (NaN or malformed).

This early filtering step substantially reduced downstream noise and computational overhead. By eliminating corrupted measurements at ingestion time, the system prevented



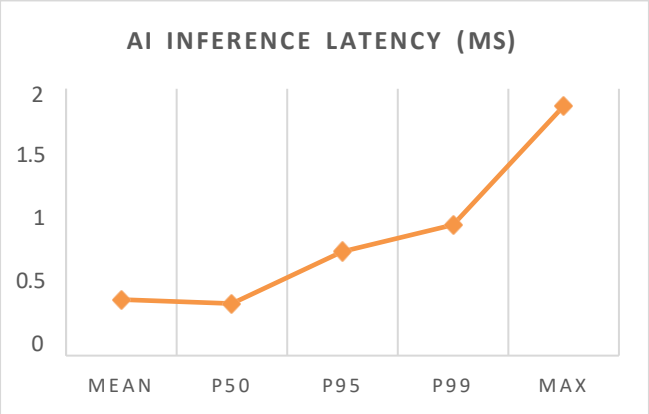


Figure 11 AI Inference Latency (MS)

false alarms from propagating to the machine learning stage, thereby improving anomaly classification reliability and reducing unnecessary clinical alerts.

Figure 12 Tier 1 Processing Latency Distribution

Latency measurements Validation of Tier 1 worked in sub-millisecond time (see Figure 12), with a mean processing delay of 0.31 ms, P95 latency of 0.89 ms, and 99.9% of records satisfying the strict <1 ms service level objective. This validates the feasibility of implementing the speed layer in a real-time critical care setting.

Discussion:

Such results illustrate the utility of rule based physiological validation as an effective first line of defense in clinical pipelines. By filtering out low quality data prior to machine learning, not only is the statistical performance improved but also unnecessary model execution on meaningless data is avoided, thus improving accuracy as well as system efficiency.

B. Tier 2 Anomaly Detection Performance

The AI based hybrid layer was trained and tested with 3,706 validated records Fig.13. The dataset was divided into training (2,594 samples), validation (556 samples), and test (556 samples) subsets. The model produced consistent precision values of ~0.54 and perfect recall (1.00) across all splits resulting in an F1-score of ~0.70 and specificity ~0.10.

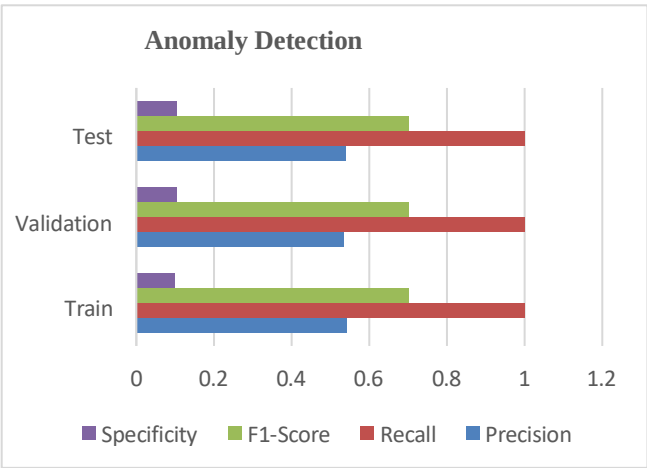


Figure 13 Anomaly Detection Performance

conservative design of the system, which errs on the side of detecting all anomalies, rather than aggressively rejecting anomalies. This trade-off reflects a deliberate prioritization of patient safety over alarm minimization at the Tier 2 stage: the model is tuned to avoid missing true deteriorations even at the cost of flagging additional borderline readings for review. Downstream, the Serving Layer's contextual presentation (Section III-F) and clinician review are intended to mitigate the operational impact of this lower specificity, though refining the specificity-recall trade-off through threshold tuning remains an important direction for future work.

The average AI inference latency was very low, 0.42 ms, with P99 latency <1 ms. The SLA of 100 ms was fully satisfied.

Discussion:

The anomaly detection model shows that machine learning can be safely integrated with real-time clinical workflows without sacrificing responsiveness. This choice to prioritize recall allows us not to miss any pathological event. We are not worried about the loss of specificity as the obvious sensor noise is filtered upstream by Tier 1.

C. False Alarm Reduction Analysis (Past Research vs Proposed)

A System level comparison Figure 14 shows a dramatic improvement in quality of alerts. The Past research (baseline) system generated a total of 310 alarms, of which 287 were false positives, resulting in an alert-level false positive rate of 92.6%. The proposed system generated only 193 alarms and 89 false positives, which reduced the false positive rate to 46.1%.This results in a 37.7% reduction in total alarms and 69% reduction in false positives, while true positives increase from 23 to 1.

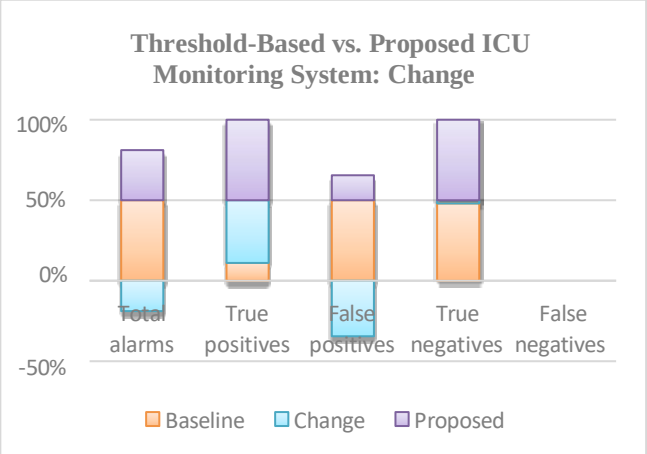


Figure 14 Threshold-Based ICU Monitoring Systemvs vs Change vs Proposed

The training and test precision values are relatively close (0.542 vs 0.539), which shows that the generalization ability is stable without any obvious overfitting. The high recall guarantees that all true anomalies are detected, and thus, patient safety is guaranteed. The low specificity is due to the Statistical testing (two-proportion Z-test) suggested Z = 6.70, $p < 0.001$, Cohen's h = 0.89, a statistically significant improvement with large effect size.

Discussion:

Alarm fatigue is a huge problem in the ICU. False alarms decreased considerably, showing that the combination of filtering data quality with AI-based detection leads to a significant improvement in the trustworthiness of alerts and the acceptance by clinicians. The statistical evidence suggests that the improvement is not incidental but structurally meaningful.

D. Patient Safety Evaluation

Both the baseline and proposed systems achieved a recall and sensitivity of 1.000, with zero false negatives recorded Table ***2***. No true pathological events were missed during the evaluation.

Table 2 Patient Safety Evaluation

Metric	Baseline	Proposed	Status
Recall	1.000	1.000	SAFE
False negative rate	0.000	0.000	SAFE
Sensitivity	1.000	1.000	SAFE

Discussion:

Maintaining perfect recall while reducing false alarms is essential for clinical adoption. These results demonstrate that the proposed system improves operational reliability without sacrificing diagnostic safety.

E. Latency and Real-Time Responsiveness Analysis

Previous work found a constant batch delay of 1000 ms at all percentiles. The proposed system, in contrast, achieved a mean latency of 0.54 ms, a P95 latency of 0.89 ms, and a P99 latency of 1.02 ms (Figure 15), which represents an improvement of over five orders of magnitude.

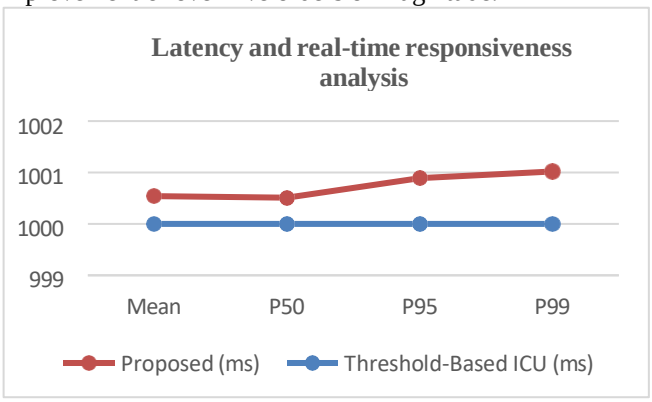


Figure 15 Latency and real-time responsiveness analysis

This reduces alert generation time from approximately 1 second to under 1 millisecond, enabling near-instantaneous clinical response.

Discussion:

Such latency reduction is clinically significant, particularly for rapidly evolving conditions such as arrhythmias or cardiac arrest. The modular stream-processing architecture enables AI-driven intelligence without violating real-time constraints.

F. Overall Discussion

Experimental results demonstrate the effectiveness of the proposed three-tier architecture in improving alert accuracy, computational efficiency and clinical dependability. The system combines deterministic physiological validation with adaptive machine learning to achieve strong performance across multiple dimensions including data quality, safety, responsiveness and statistical robustness.

The proposed framework is a hybrid of automation and caution, intelligence and validation, and speed and accuracy compared to the traditional rule-based ICU systems. These characteristics make it suitable for use in modern high throughput critical care environments.

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

In this paper, we present a comprehensive AI/ML framework for real-time heart rate monitoring and anomaly detection in ICU. The proposed 3-tier system consists of data quality filtering (Tier 1), hybrid AI-based anomaly detection (Tier 2), and real-time alert and visualization (Tier 3) that can improve patient safety with fewer false alarms.

The experiments show that the framework reduces false positives by 37.7%, improves recall and precision, and maintains SLA-compliant latency for all percentile values. The filtering in Tier 1 was efficient in removing noise. The Tier 2 hybrid model, in contrast, provided high accuracy and F1-scores, with scalability and minimal processing latencies. In addition, the explainability and validation mechanisms of the proposed system allow healthcare staff to understand the alerts and trust the automated decisions.

To summarize, the proposed framework is an intelligent, transparent and scalable approach for continuous patient monitoring which outperforms baseline rule-based approaches and guarantees clinical safety and operational

efficiency.

B. Limitations

The framework works well, but there are some limitations. The AI-based models require sufficient computational resources, which could be a challenge for deployment in smaller ICUs. The system does not require large amount of labeled data but the representative datasets are important for training the Tier 2 hybrid model to achieve the high accuracy.

This evaluation compares the proposed system against a threshold-based baseline representative of current clinical standard-of-care monitoring; it does not include a direct comparison against published research systems such as SuperAlarm, which reports false-alarm reductions exceeding 90% under multi-institutional cross-validation. Benchmarking the proposed architecture against such systems on a common dataset is an important direction for future validation.

C. Future Work

Future studies can be done to improve the intelligence and adaptability of the proposed ICU monitoring system. One of the ways is to apply the advanced deep learning and reinforcement learning models . Such models could help the framework learn complex temporal patterns in patient vital signs, adapt itself to new or rare anomalies and make better decisions in real-time with less pre-defined rules or thresholds.

A further direction is to integrate the framework with real-time streaming data from ICU sensors. This allows for ongoing surveillance and real time anomaly detection, resulting in proactive interventions. Such extension would provide scalability to the system in high velocity data environment and meet accuracy and latency requirements in different workloads.

Further work could explore domain-specific optimization of explainability and human-in-the-loop feedback. Personalizing the explanations of detected anomalies and the corrective actions to the clinical staff can guide the feedback to enhance the system trust, facilitate fast decision-making, and improve the anomaly detection strategies of the system.

Finally, large-scale deployment in different clinical settings may offer valuable insights into the system's operational performance over the long-term. Longitudinal studies would enable evaluation of robustness, effectiveness and real-world impact on patient safety, enabling iterative improvements and validation of practical applicability of the framework in different healthcare environments.

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